เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1 | Year: 2016

Subject: 394173 Mathematics III | Section: 15-16

Date: 30 November 2016 | Time: 13.00-16.00

Name ______________________________ ID ________________________ Class _______

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks, dictionary and Calculators are Not allowed.
4. This is a closed book examination.
5. Electronic communication device are **Not** allow in the examination room.
6. The examination has 8 pages (including this page), 8 questions and 1 part. The total score is 37 points.
7. Write all your answers on this examination sheet.

No. 1	No. 2	No. 3	No. 4	No. 5	No. 6	No. 7	No. 8	Total score

You are not allowed to leave the exam room during the first two hours after the beginning of the exam.

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You are not allowed to use the restroom during the exam except in case of an emergency.

1. Assume that $k-2,\ k-6,\ 2k+3$ is the geometric sequence, where $k > 0$ and $15, A, B, C, -1$ is the arithmetic sequence which $a_{10} = D$.
Determine $(A+B+C+D)^k$ (5 points)

2. Find each limit, or state that the limit does not exist and explain your reasoning.

2.1 $a_n = \left(\dfrac{8n^4 - 5n^3 + 3n - 1}{16n^4 + 3n^2}\right) + \left(\sqrt{n^2 - 4} - \sqrt{n^2 - 9}\right)$ (3 points)

2.2 $a_n = (-1)^{n+3}\dfrac{3^n + 3 \cdot 9^n + 2 \cdot 27^n}{3^{3n} \cdot 54}$ (3 points)

3. Let $\sum_{n=1}^{4}(bn^3 + 10) = 540$ (4 points)

Find the sum of first b terms of $2 \cdot 5 + 4 \cdot 6 + 6 \cdot 7 + 8 \cdot 8 + \ldots$ (Use $\sum$)

4. If the sum of first m terms of $7 + 15 + 23 + \ldots = 217$.

Find $\dfrac{3^{3m} + 3^{3m-1} + 3^{3m-2} + \ldots + 3^{16}}{3^{17}}$. (5 points)

5. Find the sum of the infinite series

$$\frac{4}{1} + \frac{4}{1+2} + \frac{4}{1+2+3} + \frac{4}{1+2+3+4} + \ldots.$$ (4 points)

6. Assume that $\vec{u} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{v} = -\hat{i} + \hat{j} + 3\hat{k}$. Find the vector has length as $|\vec{u} \cdot \vec{v}|$ which has the same direction as $\vec{u} - 3\vec{v}$? (4 points)

7. Assume that $\vec{u}$, $\vec{v}$ and $\vec{w}$ are the vectors such that $\vec{u} + \vec{v} + \vec{w} = \vec{0}$ and $|\vec{u}| = 1$, $|\vec{v}| = 4$, $|\vec{w}| = 5$. Find $3|\vec{u} - \vec{v}| - 2|\vec{v} - \vec{w}|$? (5 points)

8. Let $A(2, 0, 3)$, $B(2, 4, 5)$ and $C(1, x, 3)$ be the points, where x is a positive integer such that the area of triangle $\triangle ABC = 3$ units^2. Find $\overrightarrow{BC} \cdot \overrightarrow{AC}$? (4 points)